

ARTICLE 3

THOROUGHFARES

1. GENERAL

A. PURPOSE

1. To establish a context-based classification of streets, roads, and driveways calibrated to the Districts of this Code.
2. To require that new streets, roads, and driveways are designed in a manner that supports and reinforces the character of the District in which they are located.
3. To provide a method for classifying existing thoroughfares into the Thoroughfare Types defined herein.
4. To resolve design conflicts between vehicular and pedestrian movement in favor of the pedestrian within urban Districts, and in favor of the vehicle within rural Districts.
5. To distinguish between the form and character of a thoroughfare (regulated by this Article) and its statutory ownership and maintenance status (governed by Title 23 of the Maine Revised Statutes).
6. To absorb, calibrate, and supersede the Newcastle Road, Driveway, and Entrance Ordinance adopted November 3, 2020, which is repealed concurrent with the adoption of this Article.

B. APPLICABILITY

1. Unless otherwise specified, this Article is applicable to all streets, roads, and driveways throughout the Town of Newcastle.
2. Applies to each new street, road, driveway, and entrance that connects to a public way, and to the alteration, enlargement, reconstruction, or discontinuance of an existing street, road, driveway, or entrance on or after the effective date of this Article.
3. Applies to the classification of existing thoroughfares in accordance with Section 5 of this Article.
4. If a permit or approval is required under this Article, no earth moving or construction may occur prior to obtaining that permit or approval.

C. AUTHORITY

1. This Article is adopted under the Municipal Home Rule Powers of the Maine Constitution, Article VIII-A and 30-A MRSA Chapter 141, and Chapter 187, and under the authority granted by the Town Meeting that ratified this amendment.

D. COMPLIANCE

1. No street, road, driveway, entrance, or other improvement within the limits of the right-of-way for any public or approved private road or street may be constructed, altered, or relocated except in accordance with this Article.
2. This Article allows waivers from its standards as explicitly authorized within this Article if the Permitting Authority determines that:
 - a. Allowance of the waiver is consistent with the purpose of this Article; and
 - b. The waiver does not exceed the allowed metric variations or qualitative criteria described in the section where the waiver is authorized.

3. Allowance of a waiver under this Article is considered an integral part of the application of this Article and is not considered the granting of a variance, as that term is defined under Maine law.

E. RELATIONSHIP TO OTHER STANDARDS

1. This Article does not abrogate, annul, or otherwise interfere with any easement, covenant, or other private agreement.
2. Where the standards of this Article impose a greater restriction than required by other ordinances, regulations, resolutions, rules, easements, covenants, or agreements, the provision of this Article must apply.
3. The standards of this Article must take precedence over those of other codes, ordinances, regulations, and standards that may be in conflict with this Article, except as provided in Section 12 with respect to state-aid roads.
4. All development activity must comply with relevant Federal and State law and regulations. Where there is a conflict between this Article and the standards of a Federal or State law, Federal or State law supersedes the standards of this Article.

2. THOROUGHFARE TYPES

A. PURPOSE

1. To establish a system of named Thoroughfare Types whose standards reinforce the character of each District.
2. To enable a single classification — the Type — to serve both as a design specification for new construction and as a designation for existing thoroughfares.
3. To order the public realm on a continuum from most urban (Main Street) to most rural (Rural Road and Rural Lane), with Driveway designated as the smallest form of vehicular access.

B. APPLICABILITY

1. Every thoroughfare within the Town of Newcastle is assigned one Type from the system in this Section, in accordance with Section 5 (Classification of Existing Thoroughfares) or Section 6 (New Thoroughfares).
2. Every Driveway is regulated as a Driveway under Section 7, calibrated by the District in which the driveway is located.

C. GENERAL

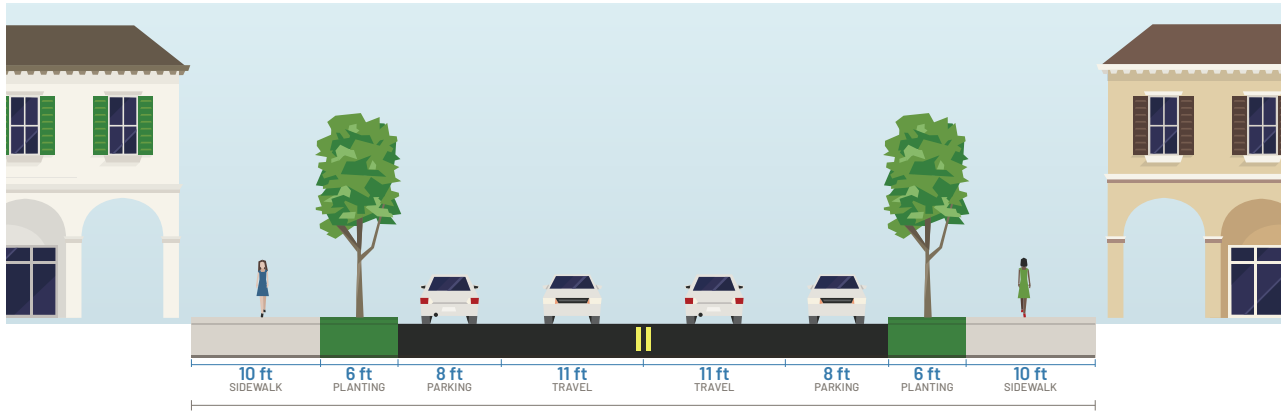
1. The Thoroughfare Types are organized in two families:
 - a. The **Street family** (S1 through S5) contains Types whose character emphasizes pedestrian comfort, defined frontages, and lower design speeds. Street Types are intended for the more urban Districts.
 - b. The **Road family** (R1 through R5) contains Types whose character emphasizes vehicular movement, working landscapes, and higher design speeds. Road Types are intended for the more rural Districts and for state-aid highway corridors.
2. The Driveway (D) is a third category, distinct from the Street and Road families, and provides private access from a Street or Road to a single lot.
3. Within Types S1 through S5 and R1 through R5, pedestrian comfort considerations are weighted as follows: in Types S1 through S4, pedestrian

comfort is a primary consideration. In Types S5 (Alley), R1, R2, and R3, pedestrian comfort is a secondary consideration but is not eliminated. In Types R4 and R5, vehicular movement is the primary consideration consistent with Maine Department of Transportation classification of the corridor.

4. The full design standards for each Type are stated on that Type's page, which follows in this Section. The sight-distance standards of Section 8 and the construction specifications of Section 11 are referenced on each Type's page and stated in full in those Sections.
5. A full page for each numbered Type (S1 through S5 and R1 through R5) follows in this Section. Each Type page states that Type's description, purpose, applicable standards, target Districts, character, and design standards, and includes an illustrative cross-section showing the representative arrangement and typical widths of the Type's right-of-way components. The cross-section illustration is representative; the design standards stated on the page govern. No Type page is provided for the Driveway (D), which is not a Thoroughfare Type and is regulated under Section 7.

MAIN STREET

S1



RIGHT-OF-WAY 66–80 ft range · shown at 73 ft

Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

The most urban Street Type — the central organizing thoroughfare of the Town Center, with continuous building streetwalls, active ground-floor uses, wide sidewalks, on-street parking both sides, and the lowest design speed in the typology.

PURPOSE

1. To establish the Town Center's principal civic and commercial thoroughfare, framed by continuous building streetwalls and active ground-floor uses.
2. To give pedestrians clear priority over vehicles, holding the lowest design speed in the typology (§1.a.4).
3. To complete the public realm with wide sidewalks, on-street parking both sides, and street trees, served from the rear by a Lane or Alley.

STANDARDS

1. Shopfront Required: a street classified S1 is a Shopfront Required street under Article 2 §4.C, subsuming the separate Shopfront Required overlay shown on the District Map.
2. A sidewalk is required both sides in D6 (§3.F.2).

TARGET DISTRICTS

- D6 Town Center
- Designated D5 Village Business segments

CHARACTER

- Buildings at the 0 ft setback line
- On-street parking buffers the sidewalk from traffic
- Planting strip and street trees both sides
- Narrow travel lanes calm traffic
- Rear-lot service via a Lane (S4) or Alley (S5)

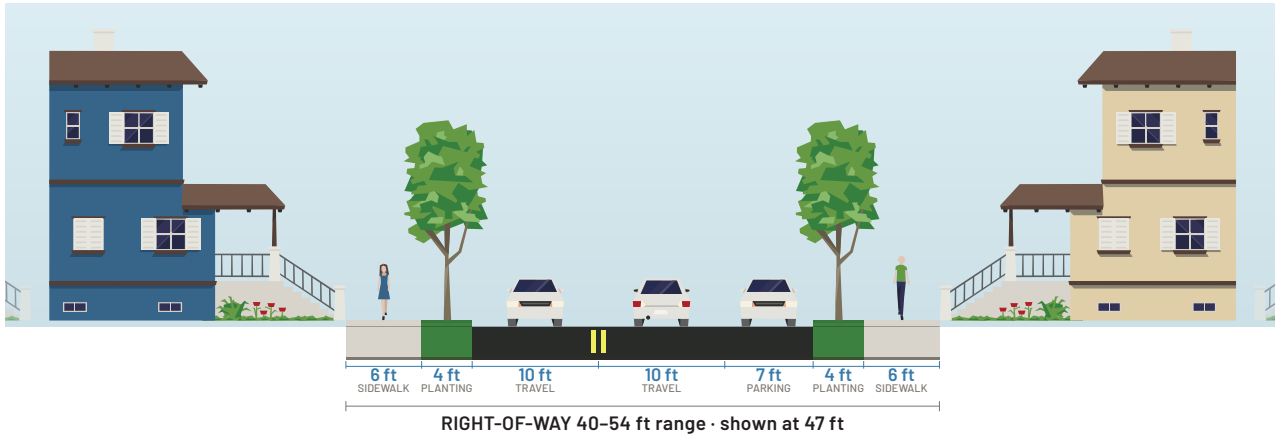
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	66–80 ft	80 ft
Traveled way	20–22 ft · 2 lanes @ 10–11 ft	20 ft · 2 @ 10 ft
On-street parking	Both sides · 7–8 ft	Both sides · 8 ft
Curb	Vertical	Vertical
Planting strip	5–8 ft, both sides	8 ft, both sides
Sidewalk	8–12 ft, both sides	12 ft, both sides
Street trees	Required	Both sides
Design speed	20 mph	20 mph
Max. grade	6%	6%
Max. block length	400 ft	400 ft
Curb return radius	10–15 ft	10 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked "where ..." are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

S2

VILLAGE STREET



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

A village-center Street with mixed-use frontages, sidewalks both sides, and on-street parking on at least one side, at a design speed suited to pedestrian-prioritized village streets.

PURPOSE

1. To support a compact, walkable village center by pairing mixed-use frontages with sidewalks and on-street parking along the travel way.
2. To resolve conflicts between vehicular and pedestrian movement in favor of the pedestrian, holding the design speed low (§1.a.4).
3. To frame the public realm with shallow setbacks, continuous street trees, and building entrances that meet the sidewalk.

STANDARDS

1. Shopfront by designation: an S2 segment may be designated Shopfront Required on the District Map and is then subject to the shopfront standards of Article 5 §12.
2. On-street parking is required where ground-floor retail is permitted (§3.F.3).

TARGET DISTRICTS

- D5 Village Business
- Edges of D6 Town Center
- Central segments of SD-Historic

CHARACTER

- Detached and attached buildings at shallow setbacks
- Primary entrances meet the sidewalk
- On-street parking at least one side; may be both
- Planting strips optional, either or both sides

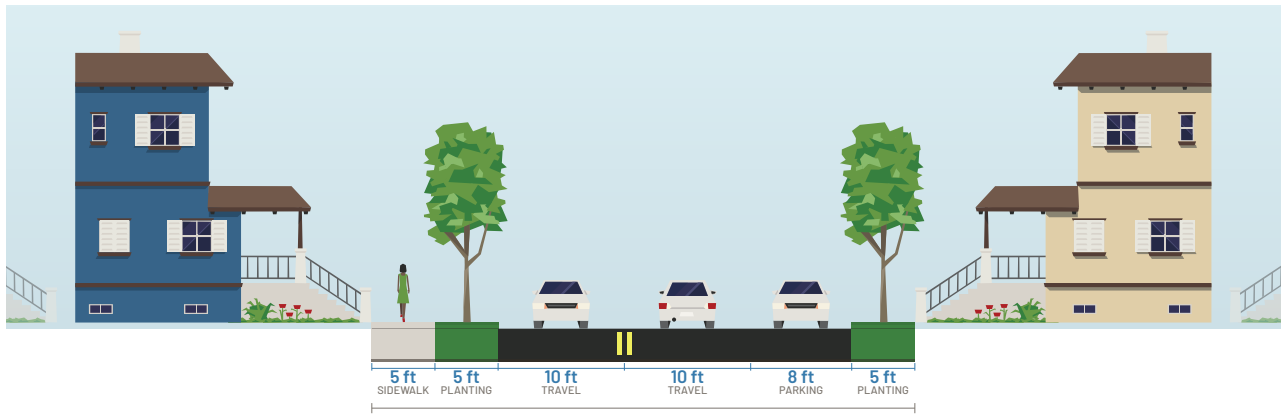
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	40–54 ft	54 ft
Traveled way	20 ft · 2 lanes @ 10 ft	20 ft · 2 @ 10 ft
On-street parking	One side · 7–8 ft	One side · 8 ft
Curb	Vertical or rolled	Vertical
Planting strip	Both sides	Both sides
Sidewalk	5 ft min, both sides	5 ft, both sides
Street trees	Required	Both sides
Design speed	25 mph	25 mph
Max. grade	8%	8%
Max. block length	500 ft	500 ft
Curb return radius	15–20 ft	15 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C): the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

NEIGHBORHOOD STREET

S3



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

The most common village Street outside the Town Center, connecting residential and small-business neighborhoods with a sidewalk and planting strip on at least one side and a residential-scale design speed.

PURPOSE

1. To connect Newcastle's residential and small-business neighborhoods at a comfortable, residential-scale design speed.
2. To provide for pedestrians with a sidewalk and planting strip on at least one side, scaled to the abutting District.
3. To accommodate on-street parking and street trees without widening the travel way beyond neighborhood scale.

STANDARDS

1. Street trees and on-street parking are provided by context under §3.F.

TARGET DISTRICTS

- D2 Neighborhood Residential
- D3 Neighborhood Business
- D4 Village Residential
- SD-Historic outside the center

CHARACTER

- Front setbacks consistent with the abutting District
- Sidewalk required on at least one side
- On-street parking permitted, one side
- Street trees encouraged in the furnishing zone

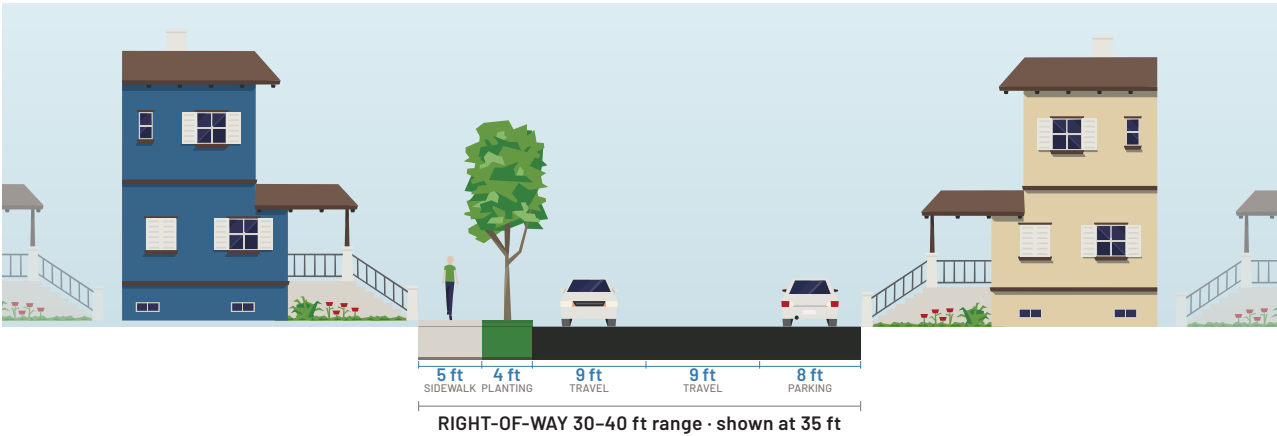
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	40-46 ft	40 ft
Traveled way	20 ft · 2 lanes @ 10 ft	20 ft · 2 @ 10 ft
On-street parking	One side · 7-8 ft	One side · 8 ft
Curb	Rolled or none	Rolled
Planting strip	4 ft min, one side	4 ft, one side
Sidewalk	5 ft min, one side	5 ft, one side
Street trees	Provided where sidewalk/planting present	Where planting built
Design speed	25 mph	25 mph
Max. grade	8%	8%
Max. block length	600 ft	600 ft
Curb return radius	15-25 ft	15 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C): the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked "where ..." are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

S4

LANE



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

A narrow, low-speed Street internal to a block that provides building frontage and access on two yield-operated travel lanes, with a required sidewalk on one side; larger than an Alley (S5), on which buildings do not front.

PURPOSE

- 1. To open block interiors to building frontage and access on a narrow, low-speed shared surface.
- 2. To provide pedestrian access with a required sidewalk on one side while holding vehicles to a yield operation.
- 3. To support compact, walkable block structure where a full Street is neither needed nor desired.

DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	30-40 ft	30 ft
Traveled way	16-20 ft · 2 lanes @ 8-10 ft	16 ft · 2 @ 8 ft
On-street parking	Optional · one side · 7-8 ft	Optional
Curb	Rolled or none	Rolled or none
Planting strip	Optional	Optional
Sidewalk	5 ft min, one side	5 ft, one side
Street trees	Provided where planting strip built	Where planting built
Design speed	15 mph	15 mph
Max. grade	10%	10%
Max. block length	n/a	n/a
Curb return radius	10-15 ft	10 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

STANDARDS

- 1. Street trees are provided where a planting strip is built (§3.F.4).

TARGET DISTRICTS

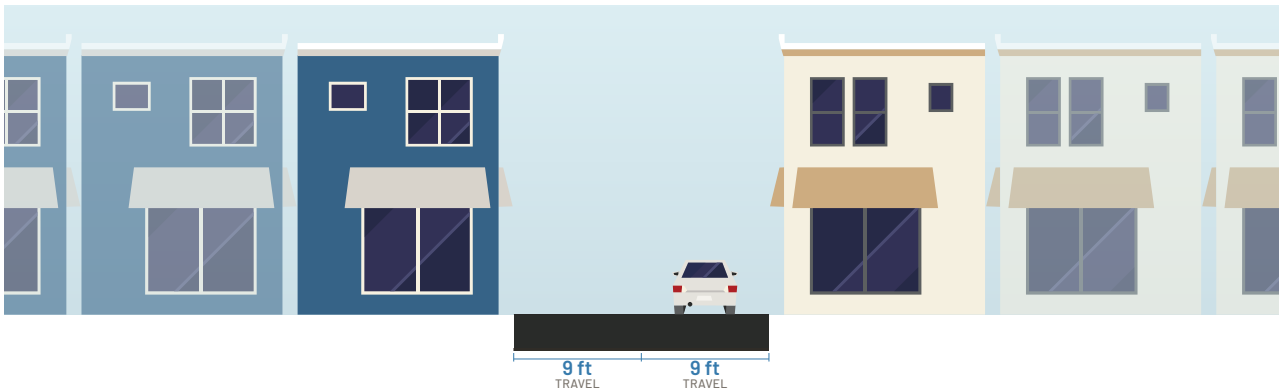
- Mid-block segments of D2 & D4
- D5 Village Business; D6 Town Center
- SD-Historic; SD-Campus

CHARACTER

- Buildings front the Lane at shallow setbacks
- Single shared surface; opposing vehicles yield
- Optional parking one side; rolled or absent curbs
- May “punch in” to open block interiors to frontage

ALLEY

S5



RIGHT-OF-WAY 16–20 ft range · shown at 18 ft

Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

A narrow service street internal to a block, providing side- and rear-lot access for parking, loading, accessory buildings, and utilities; buildings do not front on an Alley.

PURPOSE

1. To provide side- and rear-lot access for parking, loading, service, and utilities away from the primary Street frontage.
2. To keep service traffic, curb cuts, and utilities off the pedestrian-priority Streets — the only Street Type allowed for rear-lot access in D6.
3. To size the paved width and curb returns for delivery and emergency vehicles without inviting through-travel.

STANDARDS

1. An Alley carries no required pedestrian components; the widths shown are minimums.

TARGET DISTRICTS

- Internal to D3–D6
- SD-Historic; SD-Fabrication; SD-Civic
- Only rear-access type permitted in D6

CHARACTER

- No sidewalk and no on-street parking
- Narrow paved width; functionally low speed
- Tight curb returns sized for delivery & emergency vehicles
- Only Street Type allowed for rear-lot access in D6

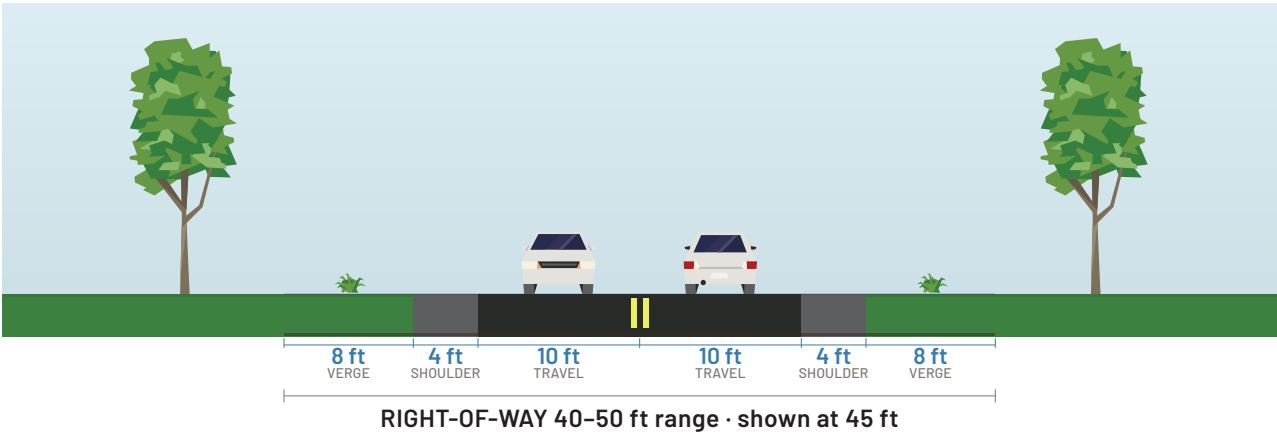
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	16–20 ft	16 ft
Traveled way	12–16 ft · 1–2 lanes @ 8–10 ft	12 ft · 1–2 @ 8 ft
On-street parking	None	None
Curb	None	None
Planting strip	None	None
Sidewalk	None	None
Street trees	Permitted	Optional
Design speed	10–15 mph	10 mph
Max. grade	10%	10%
Max. block length	n/a	n/a
Curb return radius	5–10 ft	5 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

R1

CONNECTOR ROAD



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

The most Street-like of the Road family, carrying through-travel in the rural-to-urban transition zone with shoulders rather than curbs, no continuous sidewalk, and a design speed higher than any Street Type.

PURPOSE

- 1. To carry through-travel between villages and rural areas across the rural-to-urban transition zone.
- 2. To favor the vehicle while still accommodating occasional pedestrians and cyclists on shoulders rather than sidewalks (§1.a.4).
- 3. To manage drainage and roadside development at a rural scale, with ditches, culverts, and larger setbacks than on Streets.

STANDARDS

- 1. A Road-family Type: the sidewalk and planting requirements of §3.F apply only where the segment abuts D5 or D6 frontage (§3.F.7).

TARGET DISTRICTS

- D1 Rural – D2 transition
- Secondary routes in SD-Rural Highway
- Rural connections between villages

CHARACTER

- Through-travel between villages and rural areas
- Shoulders accommodate occasional pedestrians & cyclists
- Drainage by ditches and culverts, not curb-and-gutter
- Larger building setbacks than on Streets

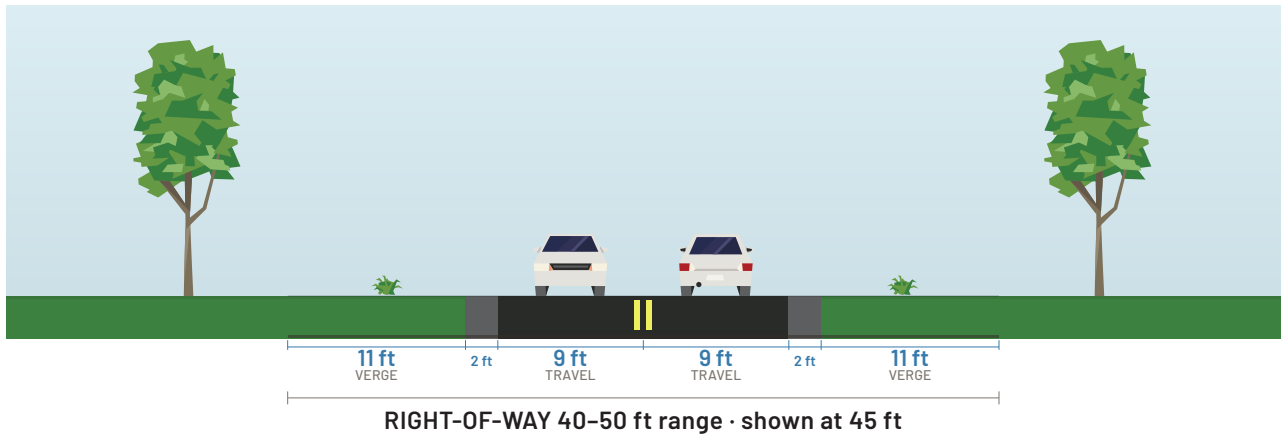
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	40–50 ft	40 ft
Traveled way	20 ft · 2 lanes @ 10 ft	20 ft · 2 @ 10 ft
On-street parking	None	None
Curb	None (shoulder)	None (shoulder)
Planting strip	Optional	Optional
Sidewalk	Optional · 5 ft	Where abuts D5/D6
Street trees	Permitted	Optional
Design speed	35 mph	35 mph
Max. grade	10%	10%
Max. block length	n/a	n/a
Curb return radius	25–35 ft	25 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved	Paved

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

RURAL ROAD

R2



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

The standard rural Road, providing access to working landscapes — farmland, forest, and conservation lands — with the narrowest standard pavement, no curbs, and drainage by open ditches.

PURPOSE

1. To provide reliable vehicular access to working rural landscapes — farm, forest, and conservation lands — with the least practical land alteration.
2. To keep the rural Road narrow and unembellished, favoring the vehicle and rural character over urban streetscape elements (§1.a.4).
3. To limit impervious surface and preserve the open, ditch-drained character of Newcastle's rural Districts.

STANDARDS

1. A rural Road-family Type: no pedestrian components are required (§3.F.7); the widths shown are minimums.

TARGET DISTRICTS

- D1 Rural
- SD-Conservation
- Rural SD-Marine segments

CHARACTER

- Paved or gravel, by segment and abutting use
- Roadside vegetation generally preserved
- Driveway entrances widely spaced (large D1 frontages)
- Rural Lane (R3) is the lower-volume variant

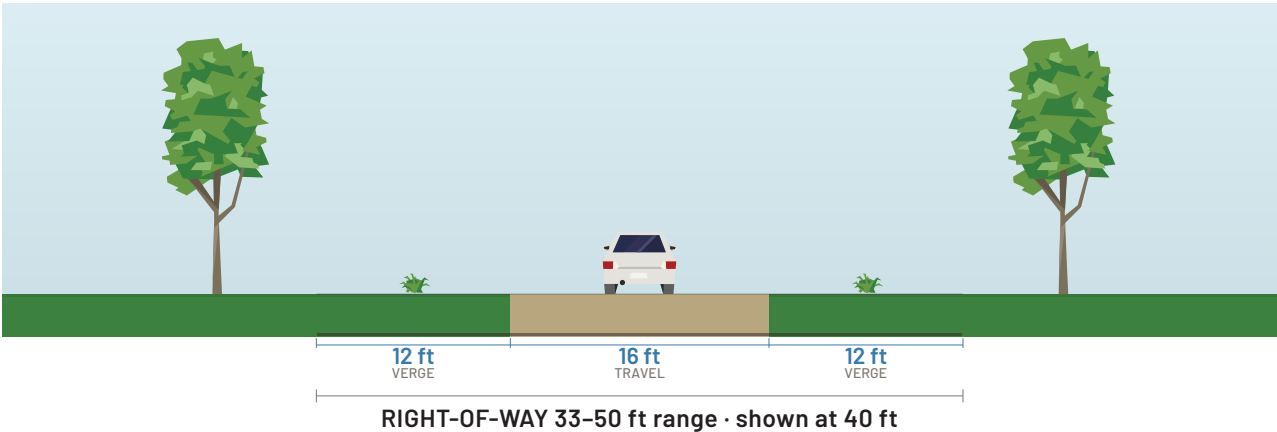
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	40–50 ft	40 ft
Traveled way	18–20 ft · 2 lanes @ 9–10 ft	18 ft · 2 @ 9 ft
On-street parking	None	None
Curb	None (shoulder)	None (shoulder)
Planting strip	None	None
Sidewalk	None	None
Street trees	Permitted	Optional
Design speed	25–35 mph	25 mph
Max. grade	12%	12%
Max. block length	n/a	n/a
Curb return radius	25–35 ft	25 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Paved or gravel	Paved or gravel

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked "where ..." are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

R3

RURAL LANE



Representative section at a typical right-of-way; widths are typical values within the ranges in the Design Standards below. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

A low-volume rural yield roadway — the rural counterpart to the Lane (S4) — carrying the fewest vehicles in the Road family (generally under 400 per day) on a single shared surface or two very narrow lanes; the conforming designation for narrow historic two-rod (33 ft) ways.

PURPOSE

- 1. To serve the lowest-volume rural ways on a single shared or very narrow surface, with the least practical land alteration.
- 2. To preserve roadside vegetation and the character of narrow two-rod ways, providing passing pull-outs in lieu of widening.
- 3. To bring existing narrow rural roads into conformance with this Code without urbanizing them.

STANDARDS

- 1. A rural Road-family Type: no pedestrian components are required (§3.F.7); passing pull-outs apply where ≤ 15 ft wide (§3.C.5).

TARGET DISTRICTS

- D1 Rural
- SD-Conservation
- Low-volume rural ways & historic two-rod (33 ft) ROWs

CHARACTER

- Gravel or paved; drainage by open ditches, no curb
- Roadside vegetation generally preserved
- Passing pull-outs at ≤ 300 ft where ≤ 15 ft wide
- Conforms existing narrow rural ways to the Code

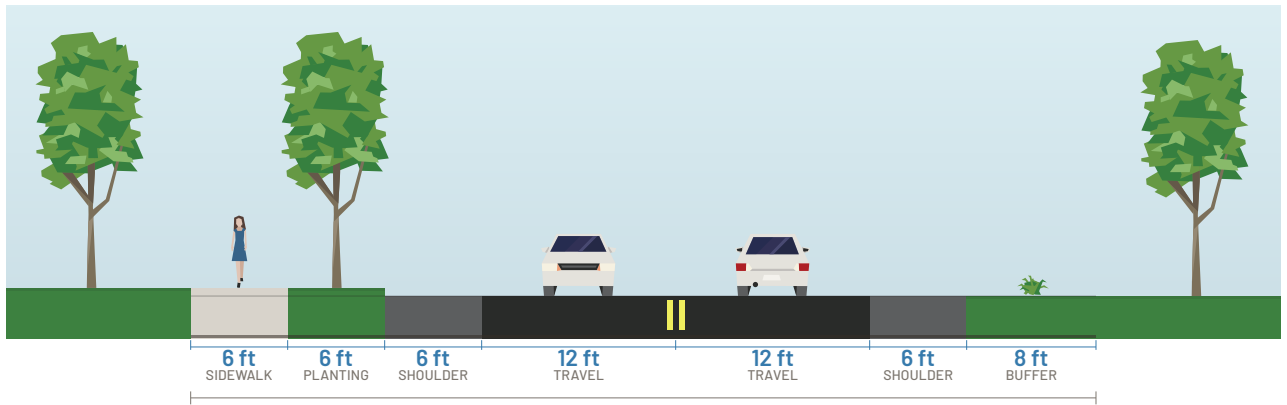
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	33–50 ft	33 ft
Traveled way	12–18 ft · 1 shared or 2 @ 8–9 ft	12 ft · 1 shared
On-street parking	None	None
Curb	None (shoulder + ditch)	None (shoulder + ditch)
Planting strip	None	None
Sidewalk	None	None
Street trees	Permitted	Optional
Design speed	15–25 mph	15 mph
Max. grade	12%	12%
Max. block length	n/a	n/a
Curb return radius	15–25 ft	15 ft
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per Table 3.3	Per Table 3.3
Surface	Gravel acceptable	Gravel

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

HIGHWAY COMMERCIAL

R4



ILLUSTRATIVE SECTION · CARTWAY & R.O.W. PER MAINEDOT

Illustrative section only. Cartway geometry and right-of-way for this Type are set by the Maine DOT (see §12); the Town governs the frontage zone, setbacks, sidewalks, and access management. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

An auto-oriented commercial corridor where the cartway is designed and maintained by MaineDOT while the Town governs the frontage zone, setbacks, sidewalks (where feasible), and access management.

PURPOSE

1. To accommodate auto-oriented commercial development along the Route 1 corridor where MaineDOT designs and maintains the cartway.
2. To secure pedestrian access and a coherent frontage where feasible, through Town control of setbacks, sidewalks, and access management (Article 2).
3. To manage curb cuts and entrances in coordination with MaineDOT (§12) so corridor safety and capacity are preserved.

STANDARDS

1. Most R4 segments are state-aid roads; the cartway follows MaineDOT geometry. See Section 12 for the MaineDOT coordination protocol.
2. Cartway per MaineDOT (§12); the Town governs frontage, sidewalks, and access management, and §3.F applies where the segment abuts D5 or D6 frontage.

TARGET DISTRICTS

- SD-Highway Commercial
- US Route 1 (most segments state-aid)

CHARACTER

- Large commercial buildings at deep front setbacks
- Sidewalks provided where feasible
- Managed curb cuts; MaineDOT §704 entrance permits
- Town governs frontage, setbacks & access (Art. 2)

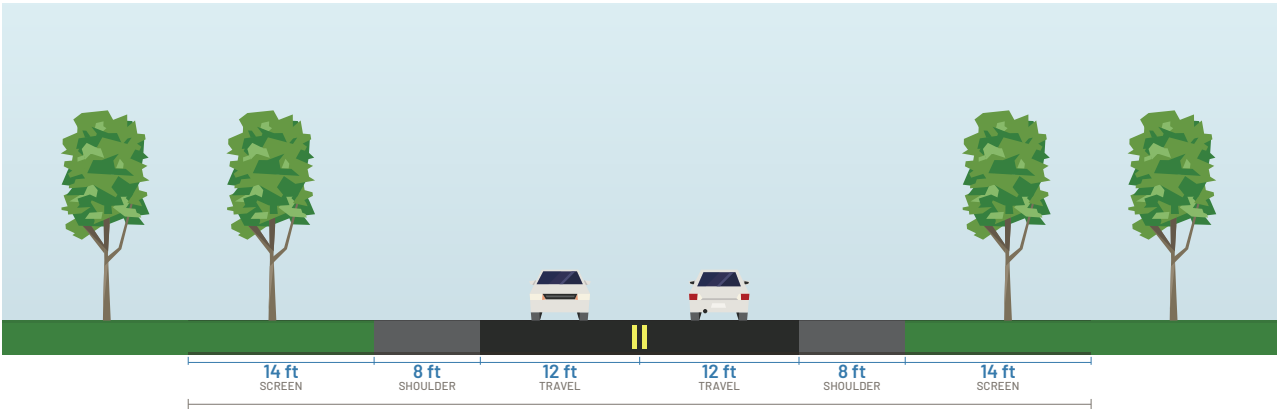
DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	Per MaineDOT	Per MaineDOT
Traveled way	Per MaineDOT	Per MaineDOT
On-street parking	Not in cartway	Not in cartway
Curb	Per MaineDOT	Per MaineDOT
Planting strip	Per District page	Per District page
Sidewalk	Where feasible	Where feasible
Street trees	Encouraged	Encouraged
Design speed	Per MaineDOT	Per MaineDOT
Max. grade	Per MaineDOT	Per MaineDOT
Max. block length	n/a	n/a
Curb return radius	Per MaineDOT	Per MaineDOT
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per MaineDOT	Per MaineDOT
Surface	Per MaineDOT	Per MaineDOT

Build to the right-hand column (Art. 3 §3.C); the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked "where ..." are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

R5

RURAL HIGHWAY



ILLUSTRATIVE SECTION · CARTWAY & R.O.W. PER MAINEDOT

Illustrative section only. Cartway geometry and right-of-way for this Type are set by the Maine DOT (see §12); the Town governs the frontage zone, setbacks, sidewalks, and access management. Cross-section illustration adapted from Streetmix (streetmix.net), © the Streetmix project, licensed CC BY-SA 4.0.

DESCRIPTION

A high-speed regional corridor through rural and conservation contexts, where MainedOT controls the cartway and the Town preserves rural character through vegetative screening, deep front setbacks, and tightly managed curb cuts.

PURPOSE

- 1. To carry high-speed regional through-travel along the Route 1 corridor on a MainedOT-controlled cartway.
- 2. To preserve rural and conservation character through vegetative screening, deep front setbacks, and tightly managed curb cuts.
- 3. To coordinate all cartway geometry and entrances with MainedOT (§12) while the Town governs the frontage and screening.

STANDARDS

- 1. All R5 segments are state-aid roads; the cartway follows MainedOT geometry. See Section 12 for the MainedOT coordination protocol.
- 2. Cartway per MainedOT (§12); rural character is preserved through vegetative screening and deep setbacks.

TARGET DISTRICTS

- SD-Rural Highway
- US Route 1 rural corridor (all segments state-aid)

CHARACTER

- Vegetative screen required between cartway and development
- Deep front setbacks measured from the centerline
- Tightly managed curb-cut spacing
- MainedOT-controlled cartway geometry (§12)

DESIGN STANDARDS

	TYPE STANDARD	BUILD TO
Right-of-way	Per MainedOT	Per MainedOT
Traveled way	Per MainedOT	Per MainedOT
On-street parking	Not in cartway	Not in cartway
Curb	Per MainedOT	Per MainedOT
Planting strip	None	None
Sidewalk	None	None
Street trees	Permitted	Optional
Design speed	Per MainedOT	Per MainedOT
Max. grade	Per MainedOT	Per MainedOT
Max. block length	n/a	n/a
Curb return radius	Per MainedOT	Per MainedOT
Sight distance	Per Table 3.2	Per Table 3.2
Pavement	Per MainedOT	Per MainedOT
Surface	Per MainedOT	Per MainedOT

Build to the right-hand column (Art. 3 §3.C): the fuller value for pedestrian and landscape components (reducible only on the written demonstration of §3.C.4), the lower value for the traffic-calming dimensions (a safety floor). Components marked “where ...” are required by context under §3.F; the Board may require fuller components within the assigned Type under §6.D.

D. DRIVEWAY (D)

1. **Description.** A Driveway is a private vehicular access way from a Street or Road to the interior of a single lot, including access to homes, parking lots, or loading docks. A Driveway is not a Street or Road and is not subject to the construction specifications of Section 11.
2. **Applicability.** A Driveway provides access to no more than two single family dwellings or one two-family dwelling on a single lot, except where a shared driveway is authorized under Section 7.B.
3. **Calibration.** Driveway standards in Section 7 are calibrated by the District in which the driveway is located.

3. THOROUGHFARE STANDARDS

A. PURPOSE

1. To state the rules of general application that govern the design standards carried on each Type's page in Section 2 — how ranges are read, how “per MaineDOT” entries operate, and the basis for the right-of-way widths and maximum grades.

B. APPLICABILITY

1. The design standards stated on the Type pages in Section 2, together with the rules of general application in this Section, apply to the design and construction of new thoroughfares under Section 6, and to the modification of existing thoroughfares under Section 14.
2. Sight distance standards (Section 8) and construction specifications (Section 11) are referenced on each Type's page but stated in full in their respective Sections.

C. GENERAL

1. Each Type's design standards are stated on that Type's page in Section 2 in two columns. The **Type standard** column states the range or choice of options the Type authorizes. The **Build to** column states the single value to be constructed — the **Build-To Standard** defined in Section 3.C.2. The rules in this Subsection govern how the two columns are read and applied.
2. The **Build-To Standard** is the value stated in the **Build to** column on the Type's page in Section 2. It is the value the applicant constructs, subject to alteration only as Section 3.C.4 provides. Where the Type standard is a single value, the Build-To Standard restates it. Where the Type standard is a range (for example, “8–12 ft”) or a choice of sides, the Build-To Standard is drawn from it as follows:
 - a. for the pedestrian and landscape components — sidewalk width, planting strip, street trees, and on-street parking — the Build-To Standard is the fuller value, and within the Street family (Types S1 through S5) these components are built; and
 - b. for the traffic-calming dimensions — travel-lane width and curb-return radius — the Build-To Standard is the lower value.
3. Where “per MaineDOT” is indicated, the Maine Department of Transportation's current geometric design standards for the applicable functional classification apply to the cartway. The Town retains authority over the frontage zone, building setbacks, sidewalks, and access management as set forth in this Article and in Article 2.
4. The Build-To Standard, together with the components stated for a Type, is the configuration to be built. It may be altered only as this Subsection provides:
 - a. **Reducing a pedestrian or landscape component.** An applicant may build a sidewalk, planting strip, street-tree provision, or on-street parking narrower than its Build-To Standard, or omit such a component, only upon a written demonstration, submitted with the application, that (i) every safety-critical standard of this Article is met, and (ii) the reduced configuration better serves the purpose of the

assigned Type in its District context under Sections 1 and 2.C. A component required under Section 3.F may not be reduced or omitted except as that Section provides.

- b. **Widening a traffic-calming dimension.** The travel-lane width and the curb-return radius build to the lower value because a narrower lane and a tighter corner slow traffic and shorten pedestrian crossings. They may be built wider than their Build-To Standard where necessary to accommodate truck, bus, or emergency-vehicle movements; they shall not be built narrower.
 - c. **Safety-critical standards are fixed.** The safety-critical standards — the traveled-way width, the sight distances of Table 3.2 and Section 8, the intersection geometry of Section 9 and the maximum grade within 75 ft of an intersection (Section 9) — are fixed for every Type and may not be reduced under this Subsection.
 - d. **Building below a minimum.** A departure under paragraph a alters the Build-To Standard within the Type's stated range and is processed as part of the application; it is not a waiver. To build below a stated minimum of the assigned Type, a waiver under Section 13.H is required.
5. For a Rural Lane (R3) whose traveled way is 15 ft or narrower, passing pull-outs sufficient for two vehicles to pass, and for the deployment of fire apparatus, shall be provided at intervals of no more than 300 ft.

D. BASIS FOR RIGHT-OF-WAY WIDTHS

1. The right-of-way widths stated on the Type pages in Section 2 are calibrated to the cross-section each Type is expected to carry — traveled way, on-street parking, planting strips, sidewalks, shoulders, and drainage — together with a margin for snow storage, dry utilities, and grading. They are not a single uniform width applied without regard to context.
2. The widths remain rooted in the traditional New England rod (1 rod = 16.5 ft): a Main Street (S1) at 66 ft is four rods, the historic town-road width of 49.5 ft is three rods, and a Rural Lane (R3) and other narrow ways descend to the two-rod (33 ft) minimum. The ranges in this Section allow a Type to default to less than the historic three-rod width where its cross-section does not require it.
3. No Maine statute prescribes a minimum right-of-way width. 30-A MRSA §4404 requires only that a subdivision not cause unreasonable highway or public-road congestion or unsafe conditions. These widths are set under the Town's home-rule authority (Section 1.C), and the safety-critical dimensions — traveled way width, sight distance (Section 8), and intersection geometry (Section 9) — are fixed independently of the right-of-way range.
4. These widths implement the 2018 Newcastle Comprehensive Plan, which found the Town's street standards “uniform” and directed that “new zoning will incorporate a range of street types and standards to accommodate local conditions and desired development goals” (*Streets & Roads: Road Standards = Land Use Goals*, p. 46). The calibrated rights-of-way further the Plan's call for “right-sized infrastructure” that reduces the Town's exposure to future road costs (*Less road*, p. 43), its observation that allowing “individual streets to become narrower ... slows traffic and increases vehicular and pedestrian safety” (*Connectivity*, p. 44), and its policies for a walkable, pedestrian-scaled Main Street and village center (*Claim Main Street*, pp. 73–74) and for the protection of rural character (*Natural & Built Landscape*, p. 116).

E. BASIS FOR MAXIMUM GRADES

1. The maximum grades stated on the Type pages in Section 2 are calibrated to each Type's context rather than to its ownership. Prior Town standards set a single maximum grade by ownership — 8% for a public road and 10% for a private road — without regard to whether the way served a pedestrian-scaled village street or a rural conduit. This Article replaces

that ownership-based figure with grades calibrated to the rural-to-urban transect.

2. The urban Street Types are held at or below the prior 8% public-road maximum: a Main Street (S1) is capped at 6% and a Village or Neighborhood Street (S2, S3) at 8%, in the interest of pedestrian comfort, accessibility, on-street parking, and winter walkability. The rural Road Types tolerate steeper grades — a Connector Road (R1) at 10% and a Rural Road or Rural Lane (R2, R3) at 12% — because they follow natural topography in low-density terrain, carry little or no pedestrian traffic, and serve the fewest lots. Permitting a modestly steeper rural grade keeps road grades as close as possible to the original topography and reduces the cut, fill, and earthwork imposed on the rural Districts (D1, D2), where this Code prioritizes minimal alteration of the landscape.
3. The steeper rural maximums relax no safety-critical standard. The 2% maximum grade within 75 ft of an intersection (Section 9), the requirement that all changes in grade be connected by vertical curves providing the minimum stopping sight distance of Table 3.2 (Section 10), and the sight-distance and intersection-geometry standards generally, are fixed independently of a Type's maximum grade and apply to every Type.
4. Grades for the Highway Commercial (R4) and Rural Highway (R5) cartways defer to MaineDOT design standards under Section 12, consistent with the status of US Route 1 as a State Highway.

F. REQUIRED COMPONENTS BY CONTEXT

1. This Subsection states the components that shall be provided on a Thoroughfare regardless of whether a component's dimension is a Build-To Standard (Section 3.C.2) or a minimum. These requirements are administered by the Code Enforcement Officer on permits the Officer issues under Section 13.C, and by the Planning Board on Subdivision and Master Plan review.
2. **Sidewalk.** A sidewalk shall be provided:
 - a. on both sides of a Street in D5 Village Business and D6 Town Center;
 - b. on at least one side of a Thoroughfare where any part of the frontage lies within ¼ mile, measured along the public way, of a school, the Town Office, the library, the post office, a public park or public landing, or a place of public assembly;
 - c. on at least one side where the segment connects to an existing sidewalk; and
 - d. on at least one side of a new Street that will serve more than [12] dwelling units.
3. **On-street parking.** On-street parking shall be provided on at least one side where the abutting frontage permits a ground-floor retail, restaurant, or personal-service use under Articles 2 and 7.
4. **Planting strip and street trees.** A planting strip with street trees shall be provided on any segment on which a sidewalk is required under Section 3.F.2, unless the Code Enforcement Officer finds in writing that a planting strip is infeasible because of a utility conflict, insufficient right-of-way remaining after the safety-critical widths are met, or ledge.
5. **Connectivity.** A new Street shall connect to any stubbed street abutting the parcel and shall provide a stub to each abutting developable parcel where a future through-connection is feasible.
6. **Trail connection.** Where an adopted Town trail, path, or sidewalk plan shows a connection across or along the parcel, the Thoroughfare shall accommodate that connection.
7. **Road-family limit.** Types R1 through R5 are exempt from Sections 3.F.2 through 3.F.4 except where the segment abuts D5 or D6 frontage.

4. OWNERSHIP CATEGORIES

A. PURPOSE

1. To distinguish the form and character of a thoroughfare, regulated as a Type under this Article, from its statutory ownership and maintenance status.
2. To recognize the four ownership categories that exist under Maine law and Town practice, and to acknowledge that any Thoroughfare Type may, in principle, exist under any of these categories.

B. APPLICABILITY

1. Every thoroughfare within the Town of Newcastle is assigned, in addition to its Type, one Ownership Category from this Section.
2. The Ownership Category of a thoroughfare is recorded on the District Map alongside its Type.

C. CATEGORIES

1. **Town Way.** A thoroughfare formally accepted by the Town under 23 MRSA §3022 by vote at Town Meeting. The Town maintains a Town Way and bears responsibility for repairs.
2. **Public Easement.** A thoroughfare over which the public holds an easement under 23 MRSA §3021, sometimes referred to historically as a "private way." The public has the right of access but the Town does not bear maintenance responsibility.
3. **Private Road.** A thoroughfare in fully private ownership over which no public access right exists. Maintenance is the responsibility of the private owner or association.
4. **State Highway.** A thoroughfare owned and maintained by the State of Maine through the Maine Department of Transportation. US Route 1 within Newcastle is a State Highway. State Highways are subject to the coordination protocol in Section 12.

D. RELATIONSHIP TO TYPE

1. The Type of a thoroughfare describes its form, character, and design standards. The Ownership Category describes its statutory ownership and maintenance status. The two designations are independent: a Main Street (S1) may be a Town Way, a Public Easement, or a State Highway; a Rural Road (R2) may be a Town Way or a Private Road.
2. Where the standards of this Article conflict with the standards or practices applicable to a particular Ownership Category — for example, with MaineDOT design standards for a State Highway, or with the terms of a Public Easement instrument — the provisions of Section 12 (Coordination with MaineDOT) and Section 4.E (below) apply.

E. PROCESS FOR CHANGING OWNERSHIP

1. The acceptance of a Private Road or Public Easement as a Town Way is a Town Meeting matter and is processed under Article 8 §20 Land Conveyance.
2. Discontinuance of a Town Way is processed under 23 MRSA §3026-A and Article 8 §27 Planning Board.
3. Changes in Ownership Category do not by themselves change the Type of a thoroughfare. The Type is set under Section 5 or Section 6 and is reviewed only under Section 14.B (Reclassification).

5. CLASSIFICATION OF EXISTING THOROUGHFARES

A. PURPOSE

1. To assign, at adoption of this Article, a Type and an Ownership Category to every existing thoroughfare within the Town of Newcastle.
2. To provide a process for reviewing and updating the classification of a thoroughfare over time.

B. APPLICABILITY

1. Applies to every named thoroughfare segment within the Town of Newcastle at the effective date of this Article.
2. A thoroughfare segment is defined as the length of a named thoroughfare between two principal intersections.

C. INVENTORY OF EXISTING THOROUGHFARES

1. The Town shall maintain an Inventory of Existing Thoroughfares (Exhibit 3.1) that, for every thoroughfare segment in Newcastle, assigns the segment a Thoroughfare Type and records descriptive information about the segment.
2. **Classification (binding).** For each segment — identified by its name and termini — the Inventory assigns a Thoroughfare Type. The Type assignment is the regulatory classification of the segment and is adopted as part of this Code. The standards applicable to a segment are the standards of its assigned Type (Section 2), together with the rules of general application in Section 3. A Type assignment may be changed only under Section 5.E (Reclassification).
3. **Recorded information (reference).** The Inventory also records the following for each segment, for reference and administration. This information is approximate, does not itself establish or limit any standard or dimension, and may be updated by the Town to reflect current conditions; where a recorded entry conflicts with the actual field condition or with a controlling standard of this Code, the actual condition or the controlling standard governs.
 - a. Ownership Category, as determined under Section 4.
 - b. Approximate right-of-way width.
 - c. Approximate traveled way width.
 - d. Adjacent District(s).
 - e. MaineDOT functional classification, where the segment is a state-aid road.
 - f. Notes on nonconformity, where the segment does not currently meet the standards of its assigned Type (Section 14).
4. **Thoroughfare Type Map.** The Thoroughfare Type Map (Exhibit 3.2) illustrates the Type assigned to each segment. It is generated from the Inventory and is provided for convenience; where the Map and the Inventory differ, the Inventory governs.
5. The initial Inventory is adopted concurrently with this Article. A change to a segment's assigned Type is processed under Section 5.E (Reclassification); recorded information under paragraph 3 may be updated by the Town without amendment of this Code.

EXHIBIT 3.1 INVENTORY OF EXISTING THOROUGHFARES

The **Type** column is the binding classification of each segment (§5.C.2). The remaining columns record reference information (§5.C.3) the Town may update without amending this Code. **DRAFT – Types auto-derived from an approximate trace of the District Map; not yet reviewed or adopted.**

THOROUGHFARE	FROM → TO	TYPE	OWNERSHIP	DISTRICT
Academy Hill	East Old County Road / Hillcrest Road → Timber Lane	 S3	Town Way	SD-Campus, D2, D1, SD-Highway Commercial, D5
Academy Hill	Kavanagh Road → Pond Road	 S3	Town Way	D4, D1
Academy Hill	Main Street / Mills Road → Route 1	 S3	Town Way	SD-Fabrication, D1
Academy Hill	Route 1 → East Old County Road / Hillcrest Road	 R2	Town Way	D1, D5
Academy Hill	Stonebridge Circle → Kavanagh Road	 S3	Town Way	D2, D1, SD-Highway Commercial
Academy Hill	Timber Lane → Stonebridge Circle	 R4	Town Way	SD-Highway Commercial
Alder Shore Road	end → end	 R2	Private Road	D1
Austin Road	Mills Road → Pond Road	 S3	Town Way	SD-Historic, D1
Averill Road	end → end	 R2	Private Road	D1
Bailey Lane	Academy Hill → end	 S3	Private Road	D2, SD-Campus
Barrol Point Road	Main Street / River Road → end	 R2	Private Road	D1
Bartlet Neck Road	end → end	 R2	Private Road	D1
Bay View Road	Academy Hill → end	 R4	Private Road	SD-Highway Commercial
Bay View Road	end → end	 R4	Private Road	SD-Highway Commercial
Bell Cove Road	end → end	 R2	Private Road	D1
Birch Road	Ridge Road → end	 R2	Private Road	D1
Blueberry Hill Road	Bunker Hill Road → end	 R2	Private Road	D1
Bluff Drive	end → High Ground Road	 R2	Private Road	D1
Bradley Shore Road	River Road → end	 R2	Private Road	SD-Conservation, D1
Briarwood Lane	Station Road → end	 R2	Private Road	D1
Brick Hill Road	Cochran Road → end	 R2	Private Road	D1, SD-Conservation
Bunker Hill Road	Jones Woods Road / Pond Road → end	 R2	Town Way	D1, D6
Camp Road	Route 1 → end	 R5	Private Road	SD-Rural Highway
Carney Road	Sheepscot Road → The Kings Highway	 R2	Private Road	D1
Castlewood Road	River Road → end	 R2	Private Road	D1
Chandler Way	end → end	 R2	Private Road	D1, D6
Chase Farm Road	West Old County Road → end	 R2	Town Way	D1
Cherry Lane	River Road → end	 R2	Private Road	D1
Cochran Road	Island Road / Station Road → end	 R2	Town Way	D1
Cole Cabin Road	Academy Hill → end	 S3	Private Road	D2
Cottage Road	end → end	 R2	Private Road	D1
Cottril Road	Academy Hill → end	 S3	Private Road	D4
Cross Street	Pump Street → Glidden Street	 S2	Town Way	D5
Cunningham Island	Island Road → end	 R2	Private Road	D1
Curlewis Lane	North Newcastle Road → end	 R2	Private Road	D1
Deer Meadow Road	Ridge Road → end	 R2	Private Road	D1
Deer Meadow Road	end → end	 R2	Private Road	D1
Depot Street	Academy Hill → end	 R2	Private Road	D1
Desperado Lane	Route 1 → end	 R4	Private Road	SD-Highway Commercial, D1
Dickenson Road	North Newcastle Road → end	 R2	Private Road	D1
Dodge Cove Lane	River Road → end	 R2	Private Road	D1, D6
Dodge Point Road	River Road → end	 R2	Private Road	SD-Conservation, D1
Doe Run	Lewis Hill Road → end	 R2	Private Road	D1
Eagle Point Road	end → end	 R2	Private Road	SD-Conservation, D1

THOROUGHFARE	FROM → TO	TYPE	OWNERSHIP	DISTRICT
East Branch Road	The Kings Highway → end	 R2	Private Road	D1
East Old County Road	Academy Hill / Hillcrest Road → end	 R2	Town Way	D1
East View Road	Mills Road → end	 S3	Private Road	D2, D1, SD-Highway Commercial
Eden Lane	Sheepscot Road → end	 R2	Private Road	D1
Elm Road	Ridge Road → end	 R2	Private Road	D1
Estey Road	Hassan Avenue → end	 R2	Private Road	D1
Evergreen Road	Ridge Road → end	 R2	Private Road	D1
Falls Road	The Kings Highway → end	 R2	Private Road	D1
Fern Birch Lane	Route 1 → end	 R5	Private Road	SD-Rural Highway, D1
Forest Road	Ridge Road → end	 R2	Private Road	D1
Gastaldo Farm Road	Academy Hill → end	 S3	Private Road	D2
Glidden Street	Cross Street → Pump Street	 S2	Town Way	D5
Glidden Street	Main Street → Cross Street	 S2	Town Way	D5
Glidden Street	Pump Street → Stewart Street	 S3	Town Way	D3
Glidden Street	Stewart Street → end	 S3	Town Way	D4, D3
Goose Cove Lane	River Road → end	 R2	Private Road	D1
Grace Lily Lane	Pond Road → end	 R2	Private Road	D1
Granite Point Road	Ridge Road → end	 R2	Private Road	D1
Gray Road	end → end	 R2	Private Road	D1
Hall Street	Academy Hill → end	 R2	Private Road	D1, D5
Happy Valley Road	Hillside Road → end	 R2	Private Road	D1
Happy Valley Road	Jones Woods Road → Hillside Road	 R2	Town Way	D1
Hassan Avenue	Ridge Road → Jones Woods Road	 R2	Town Way	D1
Hawthorn Road	Route 1 → end	 R2	Private Road	D1, SD-Rural Highway
High Ground Road	Jones Woods Road → end	 R2	Town Way	D1
High Street	Academy Hill → end	 S3	Town Way	SD-Civic, SD-Fabrication, D1
High View Road	Bunker Hill Road → end	 R2	Private Road	D1
Highland Road	North Newcastle Road → end	 R2	Private Road	D1
Hillcrest Road	Academy Hill / East Old County Road → end	 R2	Town Way	D1
Hillside Road	Jones Woods Road → end	 R2	Town Way	D1
Hodkins Road	River Road → end	 R2	Private Road	D1
Holstein Lane	Jones Woods Road → end	 R2	Private Road	D1
Hopkins Hill Road	Route 1 → end	 R2	Town Way	D1
Hunt Acres Road	Bunker Hill Road → end	 R2	Private Road	D1
Hunt Hill Road	Bunker Hill Road → end	 R2	Private Road	D1
Indian Trail	Sheepscot Road → end	 R2	Private Road	D1
Island Road	Cochran Road / Station Road → end	 R2	Town Way	D1
Jackies Way	West Hamlet Road → end	 R2	Private Road	D1
Jones Woods Road	Bunker Hill Road / Pond Road → Ridge Road	 R1	Town Way	D1
Jones Woods Road	Hassan Avenue → North Dyer Neck Road	 R1	Town Way	D1
Jones Woods Road	High Ground Road → end	 R1	Town Way	D1, D6
Jones Woods Road	Hillside Road → High Ground Road	 R1	Town Way	D1
Jones Woods Road	North Dyer Neck Road → Hillside Road	 R1	Town Way	D1
Jones Woods Road	North Newcastle Road → Hassan Avenue	 R1	Town Way	D1
Jones Woods Road	Ridge Road → North Newcastle Road	 R1	Town Way	D1
Kavanagh Road	Academy Hill → end	 R2	—	D1
Kiln Road	River Road → end	R2	Private Road	D1
Kistler Way	Pond Road → end	R2	Private Road	D1
Lake Meadow Lane	Pond Road → end	R2	Private Road	D1

THOROUGHFARES

THOROUGHFARE	FROM → TO	TYPE	OWNERSHIP	DISTRICT
Layden Lane	Sheepscot Road → end	 R2	Private Road	D1
Ledge Hill Crossing	Sheepscot Road → end	 R4	Private Road	SD-Highway Commercial, D1
Lewis Hill Road	Sheepscot Road → Lynch Road / Route 1	 R2	Town Way	D1, SD-Rural Highway
Libby Road	Pond Road → end	 R2	Private Road	D1
Liberty Street	River Road → River Road	 R2	Town Way	D1
Lincoln Lane	Pond Road → end	 R2	Private Road	D1
Little Point Road	River Road → end	 R2	Private Road	D1
Long Wharf Road	River Road → end	 R2	Private Road	D1
Lucky Lane	North Newcastle Road → end	 R2	Private Road	D1
Lynch Road	Lewis Hill Road / Route 1 → end	 R2	Town Way	D1, SD-Rural Highway
Main Street	Academy Hill / Mills Road → Pump Street	 S1	State Highway	D1
Main Street	Glidden Street → end	 S1	State Highway	D5
Main Street	Pump Street → Glidden Street	 S1	State Highway	D5
Main Street	River Road → River Road	 R1	State Highway	D1
Main Street	River Road → River Road	 R1	State Highway	D1
Main Street	River Road → Academy Hill / Mills Road	 S2	State Highway	D1
Main Street	Route 1 → Route 1	 R1	State Highway	D1
Main Street	Route 1 → end	 R1	State Highway	D1
Main Street	end → end	 R1	State Highway	D1
Main Street	end → Route 1	 R1	State Highway	D1
Main Street	end → River Road	 R1	State Highway	D1
Main Street	end → River Road	 R1	State Highway	D1
Mandys Way	North Newcastle Road → end	 R2	Private Road	D1
Maple Lane	Bunker Hill Road → end	 R2	Private Road	D1
Marsh River Run	Sheepscot Road → end	 R2	Private Road	D1
Marsh View Lane	Station Road → end	 R2	Private Road	D1
McNeil Point Road	Pond Road → end	 R2	Private Road	D1
Meadow Ridge Lane	Jones Woods Road → end	 R2	Private Road	D1
Mechanics Way	Station Road → end	 R2	Private Road	D1
Middle Way	Academy Hill → end	 S3	Private Road	SD-Campus
Milliken Island Road	Bunker Hill Road → end	 R2	Private Road	D1
Mills Road	Academy Hill / Main Street → Stewart Street	 S3	Town Way	D1, SD-Fabrication, SD-Campus
Mills Road	Austin Road → end	 R1	Town Way	SD-Conservation
Mills Road	Off To Mills Road → Old Austin Road	 S3	Town Way	D2, D1, SD-Highway Commercial, SD- Conservation
Mills Road	Old Austin Road → Austin Road	 R1	Town Way	SD-Conservation
Mills Road	On Route One Road → Route 1	 S3	Town Way	SD-Campus
Mills Road	Route 1 → Off To Mills Road	 S3	Town Way	D2, SD-Campus
Mills Road	Stewart Street → On Route One Road	 S3	Town Way	SD-Campus
Morgan Brook Lane	Jones Woods Road → end	 R2	Private Road	D1
Nob Hill Road	end → end	 R2	Private Road	D1
North Dyer Neck Road	Jones Woods Road → end	 R2	Town Way	D1, D5
North Newcastle Road	Sheepscot Road / West Old County Road → Jones Woods Road	 R2	Town Way	D1
Off To Mills Road	Mills Road → Route 1	 R4	State Highway	SD-Highway Commercial, D2, D1
Old Austin Road	Mills Road → end	 R2	Town Way	SD-Conservation
Old Field Road	end → end	 R2	Private Road	D1
On Route One Road	Mills Road → Route 1	 R4	State Highway	SD-Campus, SD- Highway Commercial, D1

THOROUGHFARE	FROM → TO	TYPE	OWNERSHIP	DISTRICT
Osprey Point Road	Route 1 → end	 R2	Private Road	SD-Conservation, D1
Oyster Lane	River Road → end	 R2	Private Road	D1
Paradise Road	North Newcastle Road → end	 R2	Private Road	D1
Perkins Point Road	River Road → end	 R2	Private Road	D1
Phils Hill Road	Jones Woods Road → end	 R2	Private Road	D1
Pine Ledge Drive	West Old County Road → end	 R2	Private Road	D1
Pinewood Farm Road	Ridge Road → end	 R2	Private Road	D1
Pleasant Street	Liberty Street → end	 S3	Town Way	D2, D1
Pond Road	Academy Hill → West Hamlet Road	 R1	Town Way	D1
Pond Road	Austin Road → Academy Hill	 R1	Town Way	D1, SD-Historic
Pond Road	West Hamlet Road → Bunker Hill Road / Jones Woods Road	 R1	Town Way	D1
Pond Road	end → Austin Road	 R2	Town Way	D1, SD-Historic
Pump Street	Cross Street → Glidden Street	 S2	Town Way	D5
Pump Street	Main Street → Cross Street	 S2	Town Way	D5
Reach Road	Sheepscot Road → end	 R2	Private Road	D1, SD-Conservation
Ridge Road	Hassan Avenue → end	 R1	Town Way	D1
Ridge Road	Jones Woods Road → Hassan Avenue	 R1	Town Way	D1
River Road	Liberty Street → Liberty Street	 R1	Town Way	D1
River Road	Liberty Street → Snead Spur	 R1	Town Way	D1
River Road	Main Street → end	 R1	Town Way	D1
River Road	Main Street → end	 R1	State Highway	D1
River Road	Main Street → end	 S2	Town Way	D1, D5
River Road	Snead Spur → end	 R1	Town Way	D1, SD-Conservation, SD-Rural Highway, D6
River Road	end → Liberty Street	 R1	Town Way	D1
River Road	end → Main Street	 R1	Town Way	D1
River Road	end → Main Street	 R1	State Highway	D1
Robinson Road	River Road → end	 R2	Town Way	D1, SD-Rural Highway
Rocky Ledge Road	North Newcastle Road → end	 R2	Private Road	D1
Route 1	Academy Hill → Main Street	 R5	State Highway	D1, SD-Civic
Route 1	Hopkins Hill Road → Snead Spur	 R5	State Highway	D1, SD-Conservation
Route 1	Lewis Hill Road / Lynch Road → Woods Island Road	 R2	State Highway	D1, SD-Rural Highway, SD-Civic
Route 1	Main Street → Main Street	 R5	State Highway	D1
Route 1	Main Street → Hopkins Hill Road	 R5	State Highway	D1
Route 1	Mills Road → Academy Hill	 R4	State Highway	SD-Campus
Route 1	Off To Mills Road → On Route One Road	 R4	State Highway	SD-Highway Commercial
Route 1	On Route One Road → Mills Road	 R4	State Highway	SD-Highway Commercial, SD- Campus, D2, D1
Route 1	Sheepscot Road → Lewis Hill Road / Lynch Road	 R5	State Highway	SD-Rural Highway, SD-Highway Commercial, D1
Route 1	Snead Spur → Sheepscot Road	 R4	State Highway	SD-Highway Commercial, D1, SD- Conservation
Route 1	Woods Island Road → end	 R5	State Highway	D1, SD-Rural Highway, SD-Civic
Route 1	end → Off To Mills Road	 R5	State Highway	SD-Conservation, D1, SD-Highway Commercial
Schraft Road	River Road → end	 R2	Private Road	D1
Shattuck Mill Road	Sheepscot Road → end	 R2	Private Road	D1
Sheepscot Road	Lewis Hill Road → North Newcastle Road / West Old County Road	 R1	Town Way	D1, SD-Civic

THOROUGHFARES

THOROUGHFARE	FROM → TO	TYPE	OWNERSHIP	DISTRICT
Sheepscot Road	North Newcastle Road / West Old County Road → The Kings Highway	 R1	Town Way	D1
Sheepscot Road	Route 1 → Lewis Hill Road	 R1	Town Way	D1, SD-Highway Commercial
Sheepscot Road	South Dyer Neck Road → end	 R1	Town Way	D1
Sheepscot Road	The Kings Highway → South Dyer Neck Road	 R1	Town Way	D1
Sheepscott River Way	end → end	 R2	Private Road	D1
Snead Spur	River Road → Route 1	 R1	Town Way	D1, SD-Rural Highway, SD-Conservation
South Dyer Neck Road	Sheepscot Road → end	 R2	Town Way	D1
Spruce Road	Ridge Road → end	 R2	Private Road	D1
Stafford Circle	Bunker Hill Road → Bunker Hill Road	 R2	Private Road	D1
Station Road	Cochran Road / Island Road → end	 R2	Town Way	D1, SD-Conservation
Station Road	end → Cochran Road / Island Road	 R2	Town Way	D1
Station Road	end → end	 R2	Town Way	D1
Stewart Street	Mills Road → Glidden Street	 R2	Town Way	D1, D3, SD-Fabrication, SD-Campus
Stonebridge Circle	Academy Hill → end	 R4	Town Way	SD-Highway Commercial, D2
Stonebridge Circle	end → end	 S3	Town Way	D1, D2
Stony Ridge Lane	West Old County Road → end	 R2	Private Road	D1, D5
Strix Hill Road	Jones Woods Road → end	 R2	Private Road	D1
Swallow Hill Road	River Road → end	 R2	Private Road	D1, SD-Rural Highway
Teague Street	Academy Hill → end	 S2	Private Road	D1, D5
The Kings Highway	Sheepscot Road → end	 R2	Town Way	D1, SD-Historic
Thistle Way	Bunker Hill Road → end	 R2	Private Road	D1
Timber Lane	Academy Hill → end	 R4	Town Way	SD-Highway Commercial
Timber Lane	end → end	 R2	Town Way	D1
Trails End Road	Cochran Road → end	 R2	Private Road	D1
Upland Way	River Road → end	 R2	Private Road	D1
Waltz Lane	Pond Road → end	 R2	Private Road	D1
Waterview Lane	River Road → end	 R2	Private Road	D1
West Hamlet Road	Pond Road → end	 R2	Town Way	D1
West Old County Road	Chase Farm Road → end	 R2	Town Way	D1, D5
West Old County Road	North Newcastle Road / Sheepscot Road → Chase Farm Road	 R2	Town Way	D1
White Road	River Road → end	 R2	Private Road	D1
Wilderness Road	North Newcastle Road → end	 R2	Private Road	D1
Woods Island Road	Route 1 → end	 R2	—	D1, SD-Civic, SD-Rural Highway

EXHIBIT 3.2 THOROUGHFARE TYPE MAP



TYPE LEGEND

S1 Main Street

S3 Neighborhood Street

R2 Rural Road

R5 Rural Highway

S2 Village Street

R1 Connector Road

R4 Highway Commercial

*Illustrative companion to the Inventory of Existing Thoroughfares (Exhibit 3.1, §5.C); generated from the Inventory and provided for convenience. The Type assignment shown is the binding classification (§5.C.2); where this Map and the Inventory differ, the Inventory governs. **DRAFT – Types auto-derived from an approximate trace of the District Map; not yet reviewed or adopted.***

D. CLASSIFICATION RUBRIC

1. The Type assigned to an existing segment is determined by applying the following rubric, in the order listed:
 - a. **Adjacent District test.** A segment whose abutting parcels are predominantly in a given District is provisionally classified as the Type targeted for that District in Section 2 (see also the District-Type cross-reference in Table 3.4).
 - b. **Built-form test.** Where the abutting built form is at variance with the Adjacent District test — for example, a segment within D2 whose abutting parcels are uniformly rural in form — the Type is adjusted to match that built form.
 - c. **Right-of-way and cartway test.** Where the existing right-of-way and cartway widths preclude one Type but are consistent with another, the Type is set to the closest match permitted by physical constraints, and the segment is recorded as nonconforming under Section 14.
 - d. **MaineDOT functional class test.** In Districts whose default Type under Table 3.4 is a Road Type — D1, SD-Conservation, SD-Highway Commercial, and SD-Rural Highway — a segment carrying a MaineDOT functional classification of Principal Arterial or Minor Arterial is presumed to be R4 (Highway Commercial) or R5 (Rural Highway) according to its abutting District, and a Collector classification is presumed to be R1 (Connector Road). In all other Districts the Adjacent District test (test a) governs regardless of functional classification: a State Highway or other MaineDOT-classified segment passing through a village or neighborhood District takes that District's targeted Type — so a Main Street that is a State Highway is classified S1 — and its cartway geometry is reconciled with State control through the Section 12 MaineDOT coordination protocol rather than by reclassifying the segment to a Road Type. A Local classification follows the Adjacent District test in every District.
 - e. **Tie-breaker.** Where the above tests produce conflicting indications, the Planning Board determines the Type after a public hearing, considering the intended long-term development pattern of the abutting District(s).

TABLE 3.4 DEFAULT TYPE BY DISTRICT (CLASSIFICATION GUIDE)

District	Default Thoroughfare Type(s)
D1 Rural	R2 Rural Road; R3 Rural Lane on low-volume segments; R1 Connector Road on transition segments
D2 Neighborhood Residential	S3 Neighborhood Street; S4 Lane on mid-block segments; R1 Connector Road on edge segments
D3 Neighborhood Business	S3 Neighborhood Street; S2 Village Street on center segments
D4 Village Residential	S3 Neighborhood Street; S4 Lane on mid-block segments; S2 Village Street on center segments
D5 Village Business	S2 Village Street; S1 Main Street on designated segments; S4 Lane internal to blocks
D6 Town Center	S1 Main Street; S4 Lane and S5 Alley internal to blocks
SD-Historic	S2 Village Street center; S3 Neighborhood Street outside center
SD-Conservation	R2 Rural Road; R3 Rural Lane on low-volume segments
SD-Highway Commercial	R4 Highway Commercial
SD-Rural Highway	R5 Rural Highway
SD-Campus	S3 Neighborhood Street; S4 Lane and S5 Alley internal
SD-Marine	S3 Neighborhood Street; R1 Connector Road on access segments
SD-Fabrication	S3 Neighborhood Street; S5 Alley internal
SD-Civic	Type follows adjacent District

E. RECLASSIFICATION

1. The Type assigned to an existing thoroughfare segment may be changed only by:
 - a. Amendment of the segment's Type assignment in the Inventory (Section 5.C) under Article 8 §21 (Zoning Amendment), processed as a Town Meeting matter; or
 - b. Reclassification under this subsection where, in connection with a proposed alteration or improvement, the abutting District has been amended or the built form has demonstrably changed such that the current Type no longer reflects either.
2. A reclassification petition may be submitted by:
 - a. The Planning Board.
 - b. The Code Enforcement Officer.
 - c. The Road Commissioner.
 - d. A property owner whose lot abuts the segment.
 - e. 10% of the number of voters who voted in the Town of Newcastle in the last gubernatorial election.
3. Reclassification petitions are reviewed by the Planning Board at a public hearing with abutter notice under Article 8 §5. The Planning Board makes a written recommendation to the Board of Selectmen, who may place the petition on the warrant of the next Town Meeting.

6. NEW THOROUGHFARES

A. PURPOSE

1. To establish the process and standards for the design and construction of new thoroughfares within the Town of Newcastle.

B. APPLICABILITY

1. Applies to:
 - a. The construction of any new thoroughfare within the Town.
 - b. Substantial reconstruction of an existing thoroughfare, where the reconstruction alters the cartway width, right-of-way width, or fundamental design speed.
2. Does not apply to:
 - a. General maintenance of an existing thoroughfare, including patching, resealing, restriping, and shoulder reshaping that does not alter the standards listed in Section 6.B.1.b.
 - b. Driveway construction or alteration, which is regulated under Section 7.

C. TYPE ASSIGNMENT AT TIME OF DESIGN

1. The Type of a new thoroughfare must be assigned at the time of design, before construction drawings are prepared.
2. The Type assignment is made by the applicant in consultation with the Code Enforcement Officer and Road Commissioner, based on the abutting District, the intended built form, and the Default Type by District guide in Table 3.4.
3. Where the new thoroughfare serves a Subdivision, the Type assignment is reviewed and approved by the Planning Board as part of the Subdivision Plan under Article 8 §12.
4. Where the new thoroughfare is a state-aid road, MaineDOT coordination is required under Section 12 prior to Type assignment.

D. DETERMINATION OF COMPONENTS AND WIDTHS

1. In approving a new Thoroughfare, the Planning Board may require components and widths drawn from within the assigned Type's stated ranges and options in Section 2 in order to serve one or more of the following objectives:
 - a. pedestrian safety and comfort consistent with Section 2.C.3;
 - b. continuity of the street, sidewalk, and block network;
 - c. the built form of the abutting District and the adopted Comprehensive Plan;
 - d. stormwater management and the establishment of tree canopy;
 - e. emergency-vehicle access; and
 - f. the supply of on-street parking.
2. A requirement imposed under Section 6.D.1 shall be supported by a written finding identifying the objective it serves and the basis for it, made as part of the Subdivision findings under Article 8 §12.F or the Master Plan findings under Article 8 §13.E.
3. The Planning Board may not require a configuration that exceeds the assigned Type. A fuller cross-section than the assigned Type provides is available only by reclassifying the segment to a higher Type under Section 5.E.
4. In Types R1 through R3, the Planning Board shall weigh the preservation of rural character and the minimization of land alteration against the objectives in Section 6.D.1.

E. DESIGN STANDARDS

1. The design of a new thoroughfare must conform to the design standards stated for the assigned Type on that Type's page in Section 2, to the rules of general application in Section 3, and to the construction specifications of Section 11, except as modified under Section 12 for state-aid roads.
2. Engineering plans must be prepared and stamped by a Maine-licensed civil engineer.
3. Plans must be reviewed and approved by the Road Commissioner prior to issuance of a Zoning Permit for thoroughfare construction.

F. CONSTRUCTION AND INSPECTION

1. Construction must be supervised by a Maine-licensed civil engineer contracted by the Town at the expense of the applicant.
2. The Road Commissioner inspects required improvements during and at the completion of construction.
3. Upon completion, the supervising engineer certifies in writing to the Road Commissioner that all required improvements have been constructed in accordance with the approved plans.

G. ACCEPTANCE AS A TOWN WAY

1. A new thoroughfare constructed under this Section is presumed to remain a Private Road unless and until it is formally accepted as a Town Way by vote at Town Meeting under 23 MRSA §3022.
2. The cost of any improvements required to bring the thoroughfare up to current Town Way acceptance standards shall not be borne by the Town.
3. The Town is not obligated to accept any privately constructed thoroughfare as a Town Way.

7. DRIVEWAYS

A. PURPOSE

1. To provide standards for vehicular access to lots.
2. To preserve the safe transition of vehicles between a Driveway and the abutting Street or Road, calibrated by the District.

B. APPLICABILITY

1. Applies to new Driveways and to the relocation of existing Driveways.
2. Does not apply to the general maintenance of an existing Driveway, including patching, resealing, and grading.

C. GENERAL

1. A Driveway provides access only from a Street or Road of a Type defined in Section 2 to the interior of a single lot.
2. A Residential Companion use may share a Driveway with the primary dwelling unit on the same lot.
3. Abutting lots may share a Driveway provided that an access easement exists between all owners and the shared Driveway complies with the entrance standards in Section 8.
4. A Driveway may pass through required front, side, and rear setbacks to access permitted parking locations.
5. One Driveway is permitted per building lot or virtual lot.
6. In D1 Rural and D2 Neighborhood Residential, more than one Driveway per lot is permitted, subject to the entrance spacing standards in Section 8.

D. DESIGN

1. All Driveways must comply with the entrance standards of Section 8 and the intersection standards of Section 9.
2. Driveways must be constructed to prevent water run-off onto the adjacent Street or Road.
 - a. In Districts D1, D2, and rural Special Districts, a Driveway may be sloped no greater than 3% above or below the adjacent cartway for the first 20 ft of the Driveway.
 - b. In Districts D3 through D6 and in urban Special Districts, the first 10 ft of a Driveway must match the slope and grade of the abutting sidewalk or, where no sidewalk is present, the abutting shoulder.
3. Where the Permitting Authority determines that a Driveway culvert is necessary, the applicant shall bear the burden of the cost for the culvert installation, which shall be constructed to the standards of Section 10.

E. CALIBRATION BY DISTRICT

1. Driveway maximum width is calibrated by District as follows:
 - a. D1, D2, R1/R2/R3 corridors: 24 ft max
 - b. D3, D4, S3 corridors: 20 ft max
 - c. D5, D6, S1/S2 corridors: 12 ft max within the frontage zone (may flare beyond the frontage zone)
 - d. SD-Highway Commercial, SD-Rural Highway: 30 ft max with MaineDOT entrance permit

8. ENTRANCES**A. PURPOSE**

1. To provide standards for the safe transition of vehicles at the point of intersection where Driveways, parking lots, and access ways meet a Street or Road.

B. APPLICABILITY

1. Applies to any new access to a lot from a Street or Road, whether utilizing a Driveway or a parking lot.

C. GENERAL

1. An entrance onto a State Highway (R4 or R5) must be approved by the Maine Department of Transportation in accordance with 23 MRSA §704. A copy of MaineDOT approval shall be submitted to the Permitting Authority at the time of application review.
2. Entrance standards apply to the first 20 ft of the Driveway, road, or parking lot that intersects the Street or Road.

D. DESIGN

1. All entrances must be designed and constructed in accordance with the latest Maine Department of Transportation's Manual for Standard Specifications.
2. No part of the entrance shall extend beyond the property lot frontage.
3. Entrance side slopes and banks shall not be steeper than a horizontal to vertical ratio of 2:1.

E. INTERSECTION ANGLE

1. Entrances shall intersect the adjacent Street or Road at a horizontal angle of 90 degrees.
 - a. Proposed horizontal angles of intersection less than 90 and greater than 75 degrees may be permitted by Special Permit.
 - b. Angles of intersection less than 75 degrees shall not be permitted.

F. LOCATION

1. All entrances shall be so located that vehicles approaching or using the entrance:
 - a. Shall be able to obtain unobstructed sight distance in both directions along the adjacent Street or Road in accordance with Table 3.2 Sight Distance.
 - b. Shall be able to maneuver safely without interference with traffic.
2. Entrances shall not be constructed within the following distances of an intersection:
 - a. 75 ft of an un-signalized intersection.
 - b. 125 ft of a signalized intersection.

G. SIGHT DISTANCE

1. Entrances shall be placed such that an exiting vehicle has an unobstructed view in both directions along the adjacent Street or Road in accordance with Table 3.2.
2. Sight distance shall be measured as follows:
 - a. From a point of origin 4 ft above ground, 10 ft from the edge of the shoulder of the adjacent Street or Road, at the horizontal center of the proposed entrance.
 - b. To a point 4 ft above ground, measured in a straight line along the Street or Road, farthest from the point of origin that can be seen without obstruction.

TABLE 3.2 SIGHT DISTANCE

Design Speed (MPH)	Sight Distance (FT)
20	155
25	200
30	250
35	305
40	360
45	425
50	570
60	645

H. CURB RADIUS

1. Entrances shall have a curb radius of 35 ft along the edge of intersection between the entrance and the adjacent traveled way, except where modified by the Type-pair geometry in Section 9.

9. INTERSECTIONS**A. PURPOSE**

1. To establish geometric standards for intersections between Thoroughfares, calibrated to the Type of each intersecting facility.

B. APPLICABILITY

1. Applies to all new intersections and to substantial reconstruction of existing intersections within the Town.
2. Intersections with State Highways (R4 or R5) are subject to MaineDOT review under Section 12.

C. GENERAL

1. The curb return radius at an intersection of two Thoroughfares is determined by the more urban Type of the two intersecting facilities, per the curb return radius stated on each Type's page in Section 2.
2. At intersections between a Street Type and a Road Type, the curb return radius is determined by the Road Type, except where the Road Type connects directly to a D5 or D6 District frontage, in which case the Street Type's radius applies.
3. Maximum grade within 75 ft of any intersection shall not exceed 2%.
4. Minimum angle of intersection between any two Thoroughfares shall be 75 degrees, with 90 degrees preferred.
5. Minimum radius along the right-of-way edge at intersections is 10 ft for Streets and 15 ft for Roads.

D. SIGHT DISTANCE AT INTERSECTIONS

1. Sight distance at an intersection shall comply with Table 3.2 for the higher design speed of the two intersecting facilities.

10. WATER MANAGEMENT

A. PURPOSE

1. To provide for the conveyance, control, and disposal of surface waters in the right-of-way of Thoroughfares, while protecting downstream water quality and aquatic habitat.

B. APPLICABILITY

1. Applies to all new Thoroughfares under Section 6 and to the substantial reconstruction of existing Thoroughfares.

C. GENERAL

1. Thoroughfares shall be designed to provide for the conveyance, control or disposal of surface waters including but not limited to the installation of ditches, culverts, catch basins, and piping systems consistent with their Type.

D. CULVERTS

1. Culverts are required for the proper movement and removal of water adjacent to Thoroughfares to ensure safe travel conditions and protect the integrity of the road surface and subsystem.
2. Culverts shall have the following standards:
 - a. Culverts must extend beyond the combined width of the traveled way and shoulders by 18 inches on both sides.
 - b. Culvert diameter shall be a minimum of 15 inches and placed at stream bed level over a prepared base.
 - c. Culverts shall be constructed to enable unobstructed fish passage in either direction.

E. DITCHES

1. Ditches shall be designed and maintained with flow lines that are 2 ft below the sub-grade breakpoint.
2. Ditches and embankments shall be seeded with erosion control landscaping.

F. GRADE CHANGES

1. Thoroughfares shall be constructed so that all changes in grade shall be connected by vertical curves in order to provide a minimum stopping sight distance per Table 3.2.

11. CONSTRUCTION STANDARDS

A. PURPOSE

1. To provide engineering specifications for the construction of Thoroughfares consistent with their Type.

B. APPLICABILITY

1. Applies to the construction of new Thoroughfares under Section 6 and to the substantial reconstruction of existing Thoroughfares under Section 14.
2. Construction of state-aid roads (R4 and R5) is governed by MaineDOT specifications per Section 12.

C. GENERAL

1. Thoroughfares shall remove stumps and rocks over 6 inches in diameter from a depth not less than 18 inches below the finished top.
2. All newly constructed Thoroughfares remain Private Roads until they are formally accepted as a Town Way per Section 6.G.

TABLE 3.3 CONSTRUCTION SPECIFICATIONS BY TYPE

Specification	S1 / S2 / S3 (Town Way)	S4 / S5 (Lane / Alley)	R1 Connector(Rural)	R2 / R3	Driveway
Construction					
Aggregate sub-base course	18"	12"	18"	12"	n/a
Crushed aggregate base course	3"	3"	3"	3"	n/a
Hot Bituminous Pavement					
Total HBP thickness	2½"	2"	2½"	2" or gravel surface	n/a
Base course	1½"	1"	1½"	1"	n/a
Surface course	1"	1"	1"	1"	n/a

D. R4 AND R5 CONSTRUCTION

1. Construction of new R4 and R5 segments and substantial reconstruction of existing R4 and R5 segments shall conform to MaineDOT specifications for the applicable functional classification.
2. Where MaineDOT specifications are less stringent than this Section for any element under Town authority (frontage zone, sidewalks, planting strips, access management), this Article controls for those elements.

12. COORDINATION WITH MAINE DOT

A. PURPOSE

1. To establish the coordination protocol between the Town of Newcastle and the Maine Department of Transportation for state-aid roads.

B. APPLICABILITY

1. Applies to all R4 (Highway Commercial) and R5 (Rural Highway) segments.
2. Applies to any entrance from a Driveway, Street, or Road onto a State Highway.

C. AUTHORITY DIVISION

1. **MaineDOT controls** the geometric design and construction of the cartway, shoulders, drainage within the cartway, and signage of State Highways, in accordance with MaineDOT's current Highway Design Guide and applicable federal standards.
2. **The Town controls**, through this Article and Article 2, the frontage zone, building setbacks, sidewalks (where present), planting strips, street trees, and access management (curb cut spacing and location) along State Highways.
3. Where a conflict arises between MaineDOT requirements and Town requirements for the same physical element, MaineDOT controls within the right-of-way and the Town controls outside the right-of-way.

D. ENTRANCE PERMITS

1. An entrance onto a State Highway requires a MaineDOT entrance permit under 23 MRSA §704 in addition to any permit required by this Article.
2. The applicant shall submit a copy of the MaineDOT entrance permit to the Permitting Authority before the Zoning Permit is issued.

E. FUNCTIONAL CLASSIFICATION

1. MaineDOT's functional classification of a segment (Principal Arterial, Minor Arterial, Major Collector, Minor Collector, Local) is independent of the Type assigned under this Article. The Type addresses form and character; the functional classification addresses transportation function for federal funding purposes.
2. The Town may consider MaineDOT functional classification as one factor in the classification rubric of Section 5.D.

13. ADMINISTRATION

A. PURPOSE

1. To describe the powers and duties of the review bodies for matters under this Article, and to coordinate Article 3 permitting with the development review framework of Article 8 (Administration).

B. APPLICABILITY

1. Applies to all permits and approvals under this Article.

C. PERMITTING AUTHORITY

1. The Code Enforcement Officer reviews and approves:
 - a. Driveway construction and relocation permits.
 - b. Entrance permits.
 - c. General maintenance permits where required.
2. The Planning Board reviews and approves:
 - a. Construction of new Thoroughfares under Section 6.
 - b. Substantial reconstruction of existing Thoroughfares under Section 14.
 - c. Reclassification petitions under Section 5.E.
3. The Road Commissioner reviews and certifies all construction-phase work.
4. The Code Enforcement Officer administers the Build-To Standards of Section 3.C and the contextual component requirements of Section 3.F on permits the Officer issues, and states any infeasibility finding under Section 3.F.4 in writing. The discretion to require fuller components and widths under Section 6.D is reserved to the Planning Board.

D. INSPECTIONS

1. There shall be at least one on-site inspection by the Permitting Authority or their representative during construction, with the assistance of the applicant's engineer. The applicant's engineer shall certify completion and compliance with the standards of this Article.

E. APPLICATION FEES

1. The application fee shall be the amount included in the Town's fee schedule.

F. PERMIT VALIDITY

1. A permit issued under this Article shall be valid for a period of twelve months from the date of original issue.

G. APPEALS

1. Whenever a person shall deem themselves aggrieved by an order made by the Permitting Authority:
 - a. The person may file an appeal to the Board of Appeals within 10 days of the date of the order.
 - b. The person shall be afforded a hearing on the matter before the Board of Appeals.

- c. Unless by their authority the aggrieved order is revoked, such order shall remain in force and be forthwith complied with by the person.

H. WAIVERS

1. The Permitting Authority may grant waivers from the dimensional standards stated for an assigned Type (Sections 2 and 3) as explicitly authorized in this Article and in accordance with Section 1.D Compliance, but may not grant waivers from the sight distance standards of Section 8.G or from the State Highway entrance permit requirement of Section 12.D.
2. A departure from a Build-To Standard under Section 3.C.4 is processed as part of the application and is not a waiver under this Subsection. A waiver under this Subsection is required only to build below a stated minimum of the assigned Type, and no waiver may be granted from the safety-critical standards listed in Section 3.C.4.

14. NONCONFORMING THOROUGHFARES

A. PURPOSE

1. To provide for the continued operation of existing Thoroughfares that do not currently meet the standards of their assigned Type, and to specify the conditions under which conformity is required.

B. APPLICABILITY

1. Applies to all Thoroughfares classified under Section 5 that do not currently meet one or more standards of their assigned Type as set forth on the Type's page in Section 2 or in Section 11 (Construction Standards).

C. CONTINUATION

1. A nonconforming Thoroughfare may continue to operate in its existing condition.
2. General maintenance — including patching, resealing, restriping, shoulder reshaping, and culvert replacement — is permitted without bringing the Thoroughfare into conformity.

D. SUBSTANTIAL RECONSTRUCTION

1. Substantial reconstruction of a nonconforming Thoroughfare requires conformity with the standards of the assigned Type, except as follows:
 - a. Where the existing right-of-way precludes a Type Standard from being met without taking additional land, the standard may be modified by the Permitting Authority to the extent necessary to fit within the existing right-of-way.
 - b. The Code Enforcement Officer or Planning Board may require the reservation of a 20 ft wide easement at the end of a dead-end road, in line with the road, to provide for continuation of pedestrian traffic or utilities.
 - c. The Permitting Authority may require the reservation of a 50 ft wide easement at the end of a dead-end road, in line with the road, to provide for continuation of the road.
2. "Substantial reconstruction" means improvements to a previously constructed Thoroughfare, including but not limited to extensions to the length or width, the rip up and replacement of sub-surface road building materials for a length or section of the Thoroughfare that goes beyond activities associated with General Maintenance, the addition of sidewalks or wholesale rebuilding of pre-existing sidewalks, or a change from one permanent road surfacing material to another such as from dirt to asphalt, asphalt to concrete, or asphalt to dirt.

E. RECLASSIFICATION TO RESOLVE NONCONFORMITY

1. Where a nonconforming Thoroughfare cannot reasonably be brought into conformity with its assigned Type, a reclassification petition may be filed under Section 5.E to assign a Type that more accurately reflects the segment's current and likely future built form.

F. ACCEPTANCE OF A NONCONFORMING PRIVATE ROAD AS A TOWN WAY

1. A Private Road that does not meet the construction standards of Section 11 for its assigned Type may not be accepted as a Town Way until the road is reconstructed to meet those standards.
2. The cost of such reconstruction shall not be borne by the Town.
3. No road shall be accepted as a Town Way without a formal vote to accept it at Annual Town Meeting.